

RESEARCH INTERESTS

Analytics for social good, public sector operations

ACADEMIC POSITIONS

Harvard Business School, Boston, MA
Technology and Operations Management Unit

2022 – Assistant Professor
2021 – 2022 Visiting Postdoctoral Fellow

Stanford University, Stanford, CA
2021 – 2022 Postdoctoral Fellow, Immigration Policy Lab

EDUCATION

2016 – 2021 **Massachusetts Institute of Technology**, Cambridge, MA
PhD in Operations Research
Advisors: Prof. Retsef Levi and Prof. Georgia Perakis
Thesis: *Healthy Food Access and Consumption: Informing Interventions Through Analytics*
Supported by the NSF Graduate Research Fellowship (2016-2019)

The Pennsylvania State University, Schreyer Honors College, University Park, PA
2013 – 2015 M.A. in Mathematics
2011 – 2015 B.A. in Mathematics
2011 – 2015 B.A. in Statistics

PRE-PRINT PAPERS

The parenthetical (A) denotes alphabetical author ordering

A Dual Perspective on Decision-Focused Learning: Scalable Training via Dual-Guided Surrogates. K. Bansak and P. Rodriguez-Diaz.

◇ Accepted to the NeurIPS 2025 MLxOR Workshop

CTRL Your Shift: Clustered Transfer Residual Learning for Many Small Datasets. G. Jain, D. Rothenhäusler, K. Bansak, and E. Paulson. Submitted.

Group Fairness in Dynamic Refugee Assignment. D. Freund, T. Lykouris, E. Paulson, B. Sturt, and W. Weng (A). Major revision at *Operations Research*.

◇ Accepted to the *24th ACM Conference on Economics and Computation* (EC '23)
◇ Accepted to the *4th annual Symposium on Foundations of Responsible Computing* (FORC '23)

PUBLISHED/ACCEPTED PAPERS

The parenthetical (A) denotes alphabetical author ordering

Heterogeneous Treatment Effects in Panel Data. R. Levi, E. Paulson, G. Perakis, and E. Zhang (A). *M&SOM* (2026, Accepted).

◇ Accepted to the 2025 Sustainable Operations SIG-Day Conference

Enhancing the benefits of dual sourcing with reverse information sharing. A. Jagmohan, P. Harsha, R. Levi, E. Paulson, and G. Perakis (A). *M&SOM* (2026, Accepted).

Dynamic Matching with Post-Allocation Service and its Application to Refugee Resettlement. K. Bansak, S. Lee, V. Manshadi, E. Paulson, and R. Niazadeh (A). *Management Science* (2026).

◇ Accepted to the 25th ACM Conference on Economics and Computation (EC '24)

◇ Accepted to the 5th Annual Symposium on Foundations of Responsible Computing (FORC '24)

◇ Accepted for Presentation at 9th Marketplace Innovation Workshop (MIW '24)

Designing Inclusive Offerings. R. Levi, E. Paulson, and G. Perakis (A). *Management Science* (2025).

Optimal Interventions for Increasing Healthy Food Consumption Among Low Income Populations. R. Levi, E. Paulson, and G. Perakis (A). *Management Science* (2025).

◇ Accepted for oral presentation, Workshop on Mechanism Design for Social Good (2020)

Public Attitudes on Performance for Algorithmic and Human Decision-Makers. K. Bansak and E. Paulson (A). *PNAS Nexus* (2024).

◇ Accepted to the 7th AAAI Conference on AI, Ethics, and Society (AIES '24)

Learning Under Random Distributional Shifts. K. Bansak, E. Paulson, and D. Rothenhäusler (A). Proceedings of the *International Conference on Artificial Intelligence and Statistics* (2024) (AISTATS '24).

Outcome-Driven Dynamic Refugee Assignment with Allocation Balancing. K. Bansak and E. Paulson (A). *Operations Research* (2024)

◇ Extended abstract appeared in the 23rd ACM Conference on Economics and Computation (EC '22)

Public Health Risks Arising from Food Supply Chains: Challenges and Opportunities. L. Chen, D. Gutierrez, R. Levi, E. Paulson, G. Perakis, N. Renegar, S. Springs (A). *Naval Research Logistics*, Special Issue on OR Models for Developmental Studies (2021).

Cooperation Can Emerge in Prisoner's Dilemma from a Multi-Species Predator Prey Replicator Dynamic. E. Paulson and C. Griffin. *Mathematical Biosciences* (2016), 178 p.56-62.

A Game Theoretic Model for Resource Allocation Among Countermeasures with Multiple Attributes. E. Paulson, I. Linkov, and J. Keisler. *European Journal of Operations Research* (2016), 252 p.610-622. DOI: 10.1016/j.ejor.2016.01.026.

Deriving and Optimally Deceptive Policy in Two-Player Iterated Games. E. Paulson and C. Griffin. Proceedings of the *American Control Conference* (2016).

Optimal Process Control of Symbolic Transfer Functions. C. Griffin and E. Paulson. Proceedings of *Feedback Computing* (2015).

Better Timing of Cyber Conflict. E. Paulson and C. Griffin. *Third ASE Conference on Cyber Security* (2014).

TEACHING MATERIALS

E. Paulson and T. Quinn. "Resilience Lab." Harvard Business School Case 625-078, March 2025.

E. Paulson, C. T. Ryan, and N. Zhang. "VOCEL(B): Powered by VOCEL." Harvard Business School Supplement 625-082, January 2025.

E. Paulson, C. T. Ryan, and N. Zhang. "VOCEL(A): Democratizing Brain Science for Early Childhood Education." Harvard Business School Case 625-081, January 2025. (Revised April 2025.)

E. Paulson and M. Toffel. "Market by Met Council: Revolutionizing Food Pantries in the Digital Age." Harvard Business School Case 624-060, May 2024.

TEACHING AT HBS

- 2022 – 2025 Technology and Operations Management (Core first-year MBA class)
- 2025 Global Colloquium on Participant-Centered Learning (2 sessions)
- 2025 Custom Executive Education Program (2 sessions)
- 2024 – 2025 Operations and Supply Chain Management, Harvard Business Analytics Program (9 sessions/year)
- 2023, 2024 Harvard Business Analytics Program Immersion (1 session per year)
- 2023 Summer Venture in Management Program (2 sessions)

OTHER TEACHING

- 2023 Harvard University, SOCIOL 1186: Refugees in Global Perspective
Guest lecture
- 2022 NYU Stern PhD Seminar Course
Guest lecture
- 2020 MIT, 15.370 - Data, Models, and Decisions
Teaching assistant for core Executive MBA class
- 2020 MIT, 15.S60 - Computing in Optimization and Statistics
3-hour session on machine learning in R for PhD students
- 2020 MIT, 15.S41 - Software Tools for Business Analytics
3-hour session on machine learning in R for MIT undergrads
- 2018 MIT, 15.734 - Intro to Operations Management
Teaching assistant for core Executive MBA class
- 2018, 2019 MIT, Microsoft Excel Training for Executive MBA students
Three 3-hour sessions each spring

- 2018 MIT, 15.731 - Risk Management
Teaching assistant for Executive MBA elective, two full-day sessions

AWARDS & HONORS

- 2025 Honorable mention, Georgia Nicholson Best Student Paper Competition, 2025, **Entrant: S. Lee**
Finalist, Service Science Best Student Paper Award, 2025, **Entrant: S. Lee**
Dynamic Matching with Post-Allocation Service and its Application to Refugee Resettlement
- 2024 1st place, Finalist, MSOM Best Student Paper Prize, 2024, **Entrant: S. Lee**
1st place, Michael H. Rothkopf Junior Researcher Paper Prize (awarded by INFORMS Auctions and Market Design Section), 2024, **Entrant: S. Lee**
Dynamic Matching with Post-Allocation Service and its Application to Refugee Resettlement
- 2021 2nd place, POMS College of Supply Chain Management Best Student Paper Competition
Enhancing the Benefits of Dual Sourcing with Reverse Information Sharing
- 2021 Finalist, POMS College of Sustainable Operations Best Student Paper Competition
Optimal Interventions for Increasing Healthy Food Consumption Among Low Income Populations
- 2019 Finalist, IBM Best Student Paper Award
Optimal Interventions for Increasing Healthy Food Consumption Among Low Income Populations
- 2016 – 2019 NSF Graduate Fellowship
- 2011 – 2015 Gerard L. Bayles Memorial Scholarship
- 2014 Kermit C. Anderson Memorial Award in Mathematics
- 2013 Mary Lister McCammon Award in Mathematics
- 2012 Merit Scholarship, Mathematics Advanced Study Semester

PRESENTATIONS

- Designing Inclusive Offerings*
- 2025 INFORMS '25
Outcome-Driven Dynamic Refugee Assignment with Allocation Balancing
- 2025 Invited seminar, University of Toronto Rotman seminar
- 2024 Invited talk, TTIC Workshop on Data-Driven Decision Processes
- 2024 Invited seminar, Wharton OID seminar
- 2024 Invited seminar, Microsoft Research New England
- 2024 Invited seminar, Northwestern Kellogg seminar
- 2024 Invited seminar, Harvard Applied Statistics workshop
- 2023 Invited seminar, Workshop on Migration and Forced Displacement
- 2023 2nd Workshop on AI & Analytics for Social Good
- 2023 Invited seminar, Northeastern, D'Amore-McKim School of Business

2023	Invited seminar, Harvard EconCS seminar
2022	EC '22
2022	MSOM '22
2021	Invited seminar, London Business School
<i>Group Fairness in Dynamic Refugee Assignment</i>	
2022	INFORMS Annual Meeting '22
<i>Short-term Outcomes for Long-term Goals in Dynamic Refugee Matching</i>	
2021	Invited seminar, Stanford Causal Science Conference
<i>Healthy Food Consumption: Empirical Analysis and Optimization Models of Government Interventions</i>	
2021	Invited seminar, Penn State College of Medicine
<i>Optimizing Group-level Food Policy Interventions</i>	
2021	Invited seminar, Stanford GSB
2021	Invited seminar, USC Marshall Business School
2021	Invited seminar, UCLA Anderson School of Business
2021	Invited seminar, Georgia Tech ISyE
2021	Invited seminar, Naval Postgraduate School
2021	Invited seminar, Harvard Business School
2021	Invited seminar, Columbia Business School
2020	Invited seminar, UBC Sauder School of Business
2020	Invited seminar, University of Minnesota ISyE
2020	Invited seminar, Boston College Carroll School of Management
2020	Invited seminar, UW Madison Wisconsin School of Business
2020	Invited seminar, UC Berkeley Haas School of Business
2020	Invited seminar, Dartmouth Tuck School of Business
2020	Invited seminar, MIT Sloan School of Management
<i>Reverse Information Sharing: Reducing Waste and Costs in Supply Chains with Yield Uncertainty</i>	
2021	POMS '21
2020	INFORMS Annual Meeting '20
<i>Fair Group-level Intervention Bundles</i>	
2021	INFORMS Annual Meeting '21
2020	INFORMS Annual Meeting '20
<i>Optimal Interventions for Healthy Food Consumption Among Low Income Households</i>	
2021	POMS '21
2020	Kellogg-Wharton OM Workshop
2020	Workshop on Mechanism Design for Social Good '20
2019	INFORMS Annual Meeting '19
2019	Invited seminar, MIT Sloan Health Systems Innovation Seminar Series
2018	INFORMS Annual Meeting '18
<i>Impact of Access and Value on Fresh Food Consumption: Policy Implications</i>	
2018	MSOM '18
2017	INFORMS '17

*Choosing What to Protect, and How: Resource Allocation Among Countermeasures
With Multiple Attributes*
2014 INFORMS '14

INDUSTRY EXPERIENCE

2019 – 2019 **IBM Blockchain Solutions**, Yorktown Heights, NY
Research Intern

2015 – 2016 **Booz Allen Hamilton**, Annapolis Junction, MD
Data Scientist

Summer **Bates White**, Washington, D.C.
2014 Summer consultant

Summer **U.S. Army Engineer Research and Development Center**, Concord, MA
2013 Research intern, DHS HS-STEM Summer Internship program

PROFESSIONAL SERVICE

Ad-hoc Reviewer for *Management Science*, *M&SOM*, *Operations Research*, *POM*

2024 Moderator for the 33rd Women in Business Conference at HBS

2024 – 2025 Program Committee, ACM EC Conference

2022 – 2024 Area Chair, ACM EAAMO Conference

2023 – 2024 Session Chair, INFORMS Annual Conference

2023 Program Committee, AAAI/ACM AIES

2024 Judge for the POMS College of Sustainable Operations Student Paper Competition

2021, 2025 Program Committee, ACM EAAMO Conference

2021, 2024 Judge for the INFORMS Public Sector OR Best Paper Award